Week 5 Project Documentation

Amazon Reviews Sentiment Analysis Dashboard

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Introduction

This project implements a Sentiment Analysis Dashboard using Streamlit. The dashboard predicts whether an Amazon customer review is positive or negative using a trained machine learning model.

Objective

Develop an end-to-end NLP application including data preprocessing, TF-IDF vectorization, machine learning model training, model saving, and deployment with Streamlit.

Dataset

The Amazon Reviews dataset contains customer reviews and sentiment labels. Reviews were cleaned before feature extraction and model training.

Technologies

Python, Pandas, Scikit-learn, Joblib, Streamlit, Matplotlib, and WordCloud.

Methodology

The workflow includes data cleaning, TF-IDF feature extraction, model training, evaluation, saving the model, and deploying an interactive dashboard.

Figure 1: Streamlit Dashboard Home Page

🛍️ Amazon Reviews Sentiment Analysis Dashboard

Welcome

This dashboard predicts whether an Amazon customer review is **Positive** or **Negative** using Machine Learning.

Dataset

- Amazon Alexa Reviews Dataset

Models Used

- Multinomial Naive Bayes
- Logistic Regression

Use the sidebar to navigate through the dashboard.

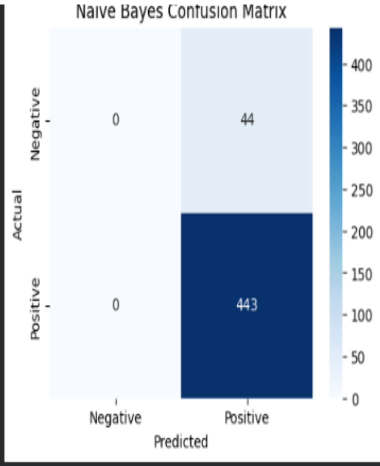
Figure 2: Sentiment Prediction Code

app.py 2 ●

app.py > ...

```
        else:

            cleaned_review = clean_text(review)

            review_vector = vectorizer.transform([cleaned_review])

            prediction = model.predict(review_vector)[0]

            confidence = model.predict_proba(review_vector).max()

            if prediction == 1:
                st.success("😊 Positive Review")
            else:
                st.error("☹️ Negative Review")

            st.write(f"### Confidence Score: {confidence:.2%}")S
```

Figure 3: Model Evaluation (Confusion Matrix)



Observation

The confusion matrix shows that the Naive Bayes model correctly classified most positive reviews but failed to identify negative reviews. This happened because the dataset is highly imbalanced, with significantly more positive reviews than negative ones. Therefore, another classification model will be trained and compared to achieve better performance.

Step 11: Train Model 2 (Logistic Regression)

Results

The application successfully predicts customer sentiment and provides an easy-to-use web interface. The trained model and vectorizer are loaded using Joblib, and predictions are displayed with confidence scores.

Conclusion

The project demonstrates practical NLP skills, machine learning model deployment, GitHub project management, and Streamlit dashboard development.